TM Forum Specification

Resource Qualification User Guide

TMF761

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Notice

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Introduction

The following document is the specification of the REST API for Resource Qualification. It includes the model definition as well as all available operations. Possible actions are retrieving CheckResourceQualification and QueryResourceQualification resources.

The TMF761 Resource Qualification API efficiently provides and recommends resource instances within networked environments based on client or front-end system inputs. Its functionality is read-only, meaning it does not configure, update, or modify resource instances. Instead, it focuses on reducing operational inefficiencies by checking or querying resource instances. It manages CheckResourceQualification and QueryResourceQualification resources:

- A CheckResourceQualification is a resource used to verify the eligibility of a configured resource (or a list of resources). A qualification result is provided in response, and requesters may ask for the reason for the result (availability or unavailability) and/or alternate proposals.
- A QueryResourceQualification is a resource used to retrieve a list of eligible resources based on search criteria provided by the requester. It returns an optimal list of resource instances including physical resources and logical resources. Based on industry practices, this API primarily suggests the following logical resource categories:
  - Transport Resources: Resources related to connectivity and data transfer, such as time slots for transmission circuits or bandwidth for IP links.
  - Cloud Resource Quotas: Resources governed by capacity or license specifications, such as compute resources (e.g., OpenStack Flavors defining CPU cores, memory, storage) or license quotas.

TMF761 Resource Qualification API performs the following operations on CheckResourceQualification resource: Retrieval of a CheckResourceQualification or a collection of CheckResourceQualification(s) depending on filter criteria TMF761 Resource Qualification API performs the following operations on QueryResourceQualification resource: Retrieval of a QueryResourceQualification or a collection of QueryResourceQualification(s) depending on filter criteria The following assumptions were considered in the development of this document:

- The Resource Qualification API is read-only and shall not support configuration, update, or modification of resource instances.
- The API relies on inputs from client or front-end systems to perform resource checking and querying.
- For QueryResourceQualification operations, priority should be given to querying the

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recommended logical resource categories (Transport Resources and Cloud Resource Quotas) as specified in industry practices.

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Sample Use Cases

To provide some user-cases

CheckResourceQualification

- Check if the existing OBD resource can support 1000 Mbps internet access for the user. The process includes verifying whether the upstream central office connected to this OBD has an OLT device with a 1000 Mbps downstream port.

> This user case happens when a personal customer would like to update the 1000 Mbps internet access. The CSP need to check whether the upstream device could support the 1000 Mbps brandwidth. In this case, the CSP may check the OBD resource or OLT resource as input.

- Check if a requested cloud resource (e.g., 8-core, 16GB server) is available in the nearest data center based on the user's location
- If the primary requested resource is exhausted, the system checks and returns alternative options (e.g., resources in a neighboring data center like Dongguan or Shenzhen)

QueryResourceQualification

- For an existing demand for 1000 Mbps internet access, query the nearest OBD (Optical Distribution Box) to the incoming address. Also, verify if the upstream central office connected to this OBD has an OLT (Optical Line Terminal) device with a 1000 Mbps downstream port. If not, search for another OBD and repeat the process. List all OBDs that can support the 1000 Mbps internet access.
- Retrieve a list of available cloud server specifications in the closest geographically eligible data center(s) when the local data center lacks capacity
- Retrieve optimal resource allocation options by cross-referencing user location (hierarchical address) with infrastructure coverage and resource availability across multiple sites

Comparison

Difference between Query and Check:

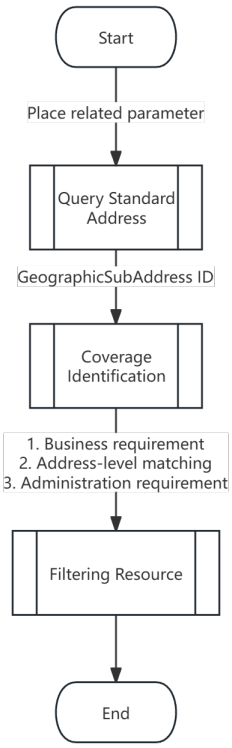
- Query focus: Retrieves a list of eligible resource instances. The Resource IDs are provided in the response payload to identify available physical or logical resources that

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match the criteria.

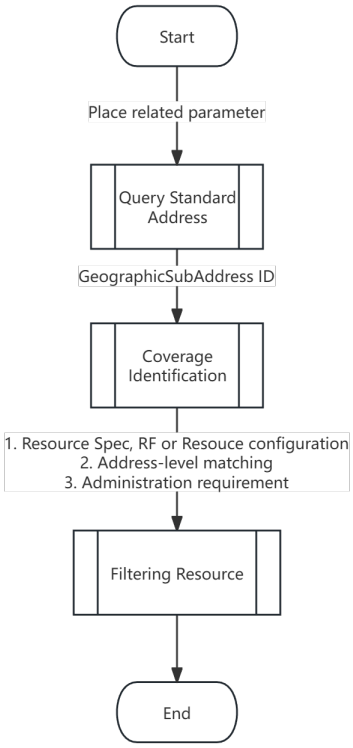
- Check focus: Checks the input resource configure including Place whether matches the network runtime data.

Example flow for resource qualification query



Example flow for resource qualification check

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 TMF630.

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

QueryResourceQualification resource

QueryResourceQualification is used to retrieve a list of resource that are technically available in the context of the interaction (place, party, resource characteristics, ...).

Resource model

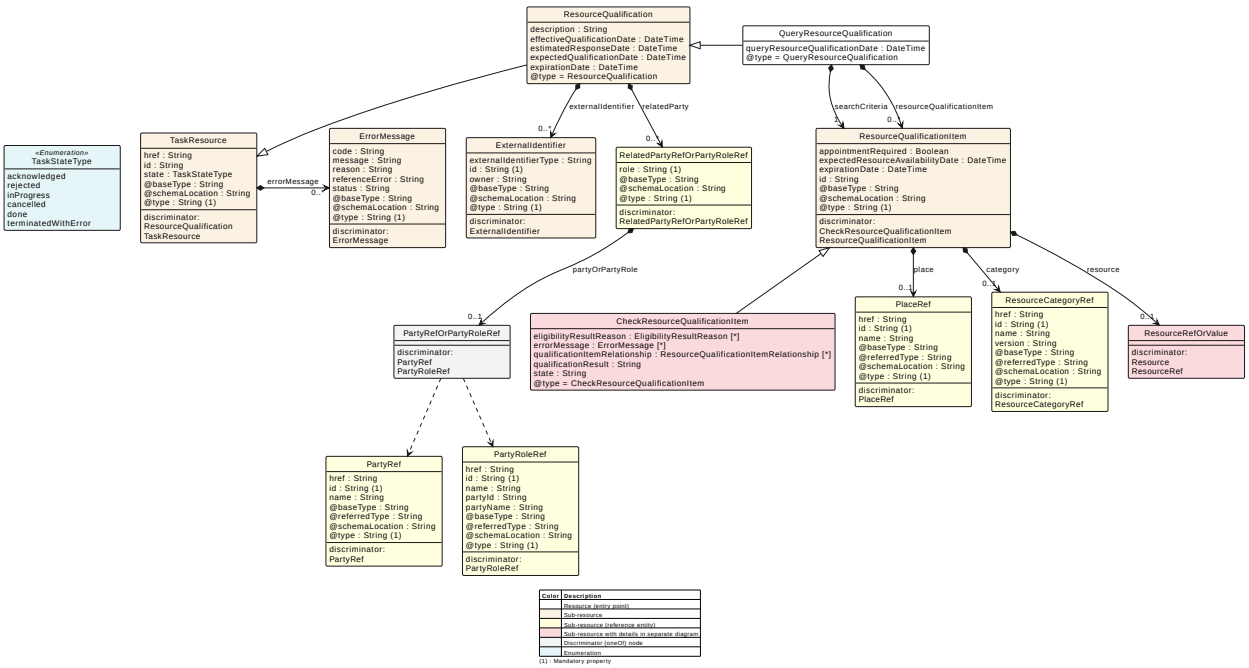


Figure 1 - QueryResourceQualification

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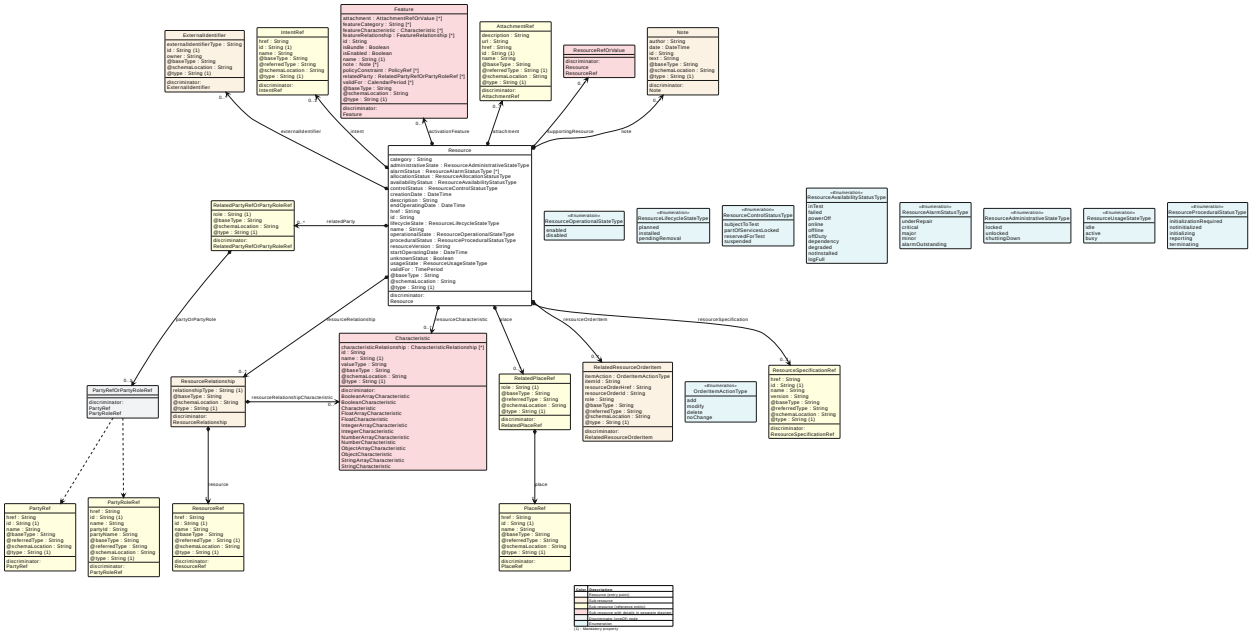


Figure 2 - Resource

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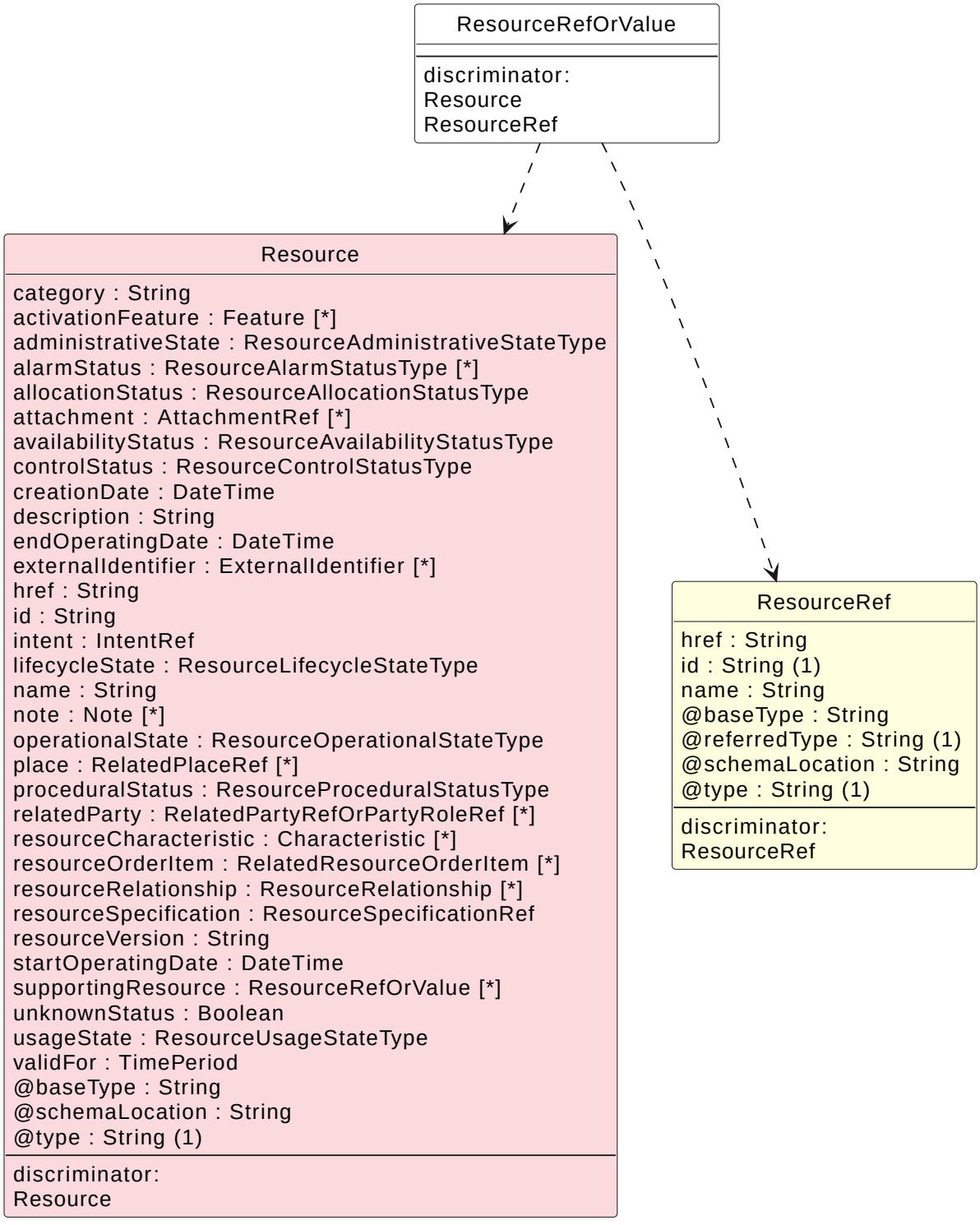


Figure 3 - ResourceRefOrValue

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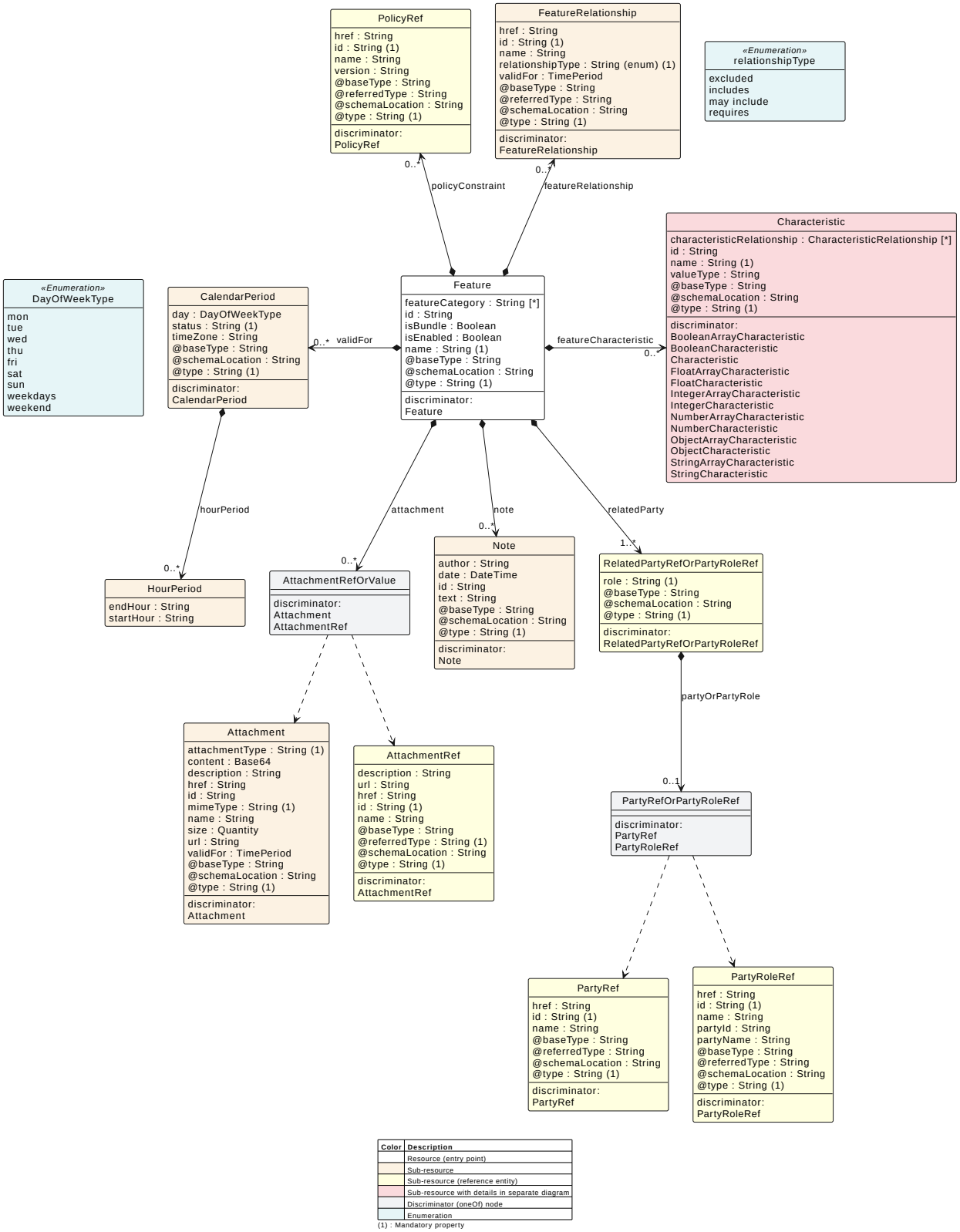


Figure 4 - Feature

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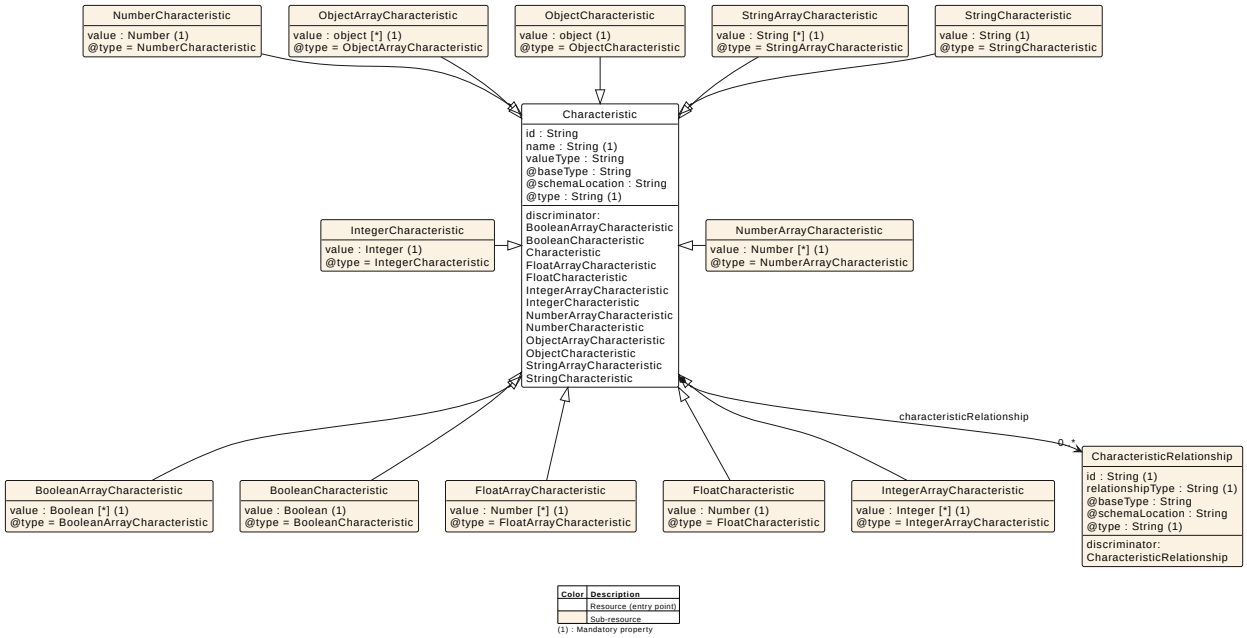


Figure 5 - Characteristic

Field descriptions

QueryResourceQualification fields

description	A String. Description of the resourceQualification.
effectiveQualificationDate	A DateTime. Effective date to resourceQualification completion.
errorMessage	An ErrorMessage [0..*]. A list of error message contains the error that caused the termination of the task.
estimatedResponseDate	A DateTime. Date when the requester expect to provide an answer for the qualification request.
expectedQualificationDate	A DateTime. A date (DateTime). Deadline date when the requester expected a qualification answer.
expirationDate	A DateTime. Date when the qualification response expires.
externalIdentifier	An ExternalIdentifier [0..*]. A list of external identifiers associated with this entity.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
queryResourceQualificationDate	A DateTime. Date when the resourceQualification was submitted.

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relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of related party references, defines party or party role linked to this request.
resourceQualificationItem	A ResourceQualificationItem [0..*]. A list of resource qualification items.
searchCriteria	A ResourceQualificationItem [1]. A ResourceQualificationItem relates to a specific resource being checked in a qualification operation.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ErrorMessage sub-resource fields

Represents an Error.

code	A String. Error code.
message	A String. More details and corrective actions related to the error.
reason	A String. Explanation of the reason for the error.
referenceError	A String. URI of documentation describing the error.
status	A String. Error code extension like sys-ABC-2001.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String [1]. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String [1]. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

A Party reference.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRef

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRoleRef

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ResourceQualificationItem sub-resource fields

A ResourceQualificationItem relates to a specific resource being checked in a qualification operation.

appointmentRequired	A Boolean. Indicates whether an appointment is required or not.
category	A ResourceCategoryRef. The (resource) category resource is used to group resource candidates in logical containers. Categories can contain other categories.
expectedResourceAvailabilityDate	A DateTime. Date when the requester looks for resource availability.
expirationDate	A DateTime. Date when the qualification item response expires.
id	A String. Locally unique ID of the of the Resource Qualification Item.
place	A PlaceRef. A reference to a physical Place.
resource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ResourceQualificationItem can be instantiated as * CheckResourceQualificationItem
value	This property is present in subclasses

ResourceCategoryRef sub-resource fields

The (resource) category resource is used to group resource candidates in logical containers. Categories can contain other categories.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.

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name	A String. Name of the referred entity.
version	A String. Category version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

A reference to a physical Place.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The specific sub-type of Place being referred to (such as a `GeographicAddress`).
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Resource sub-resource fields

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources (e.g. TPE, EQUIPMENT) in the inventory.

activationFeature	A Feature [0..*]. Configuration features; see TMF664 Resource Function Activation API.

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administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType [0..*]. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef [0..*]. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.

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externalIdentifier	An ExternalIdentifier [0..*]. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note [0..*]. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled

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place	A RelatedPlaceRef [0..*]. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating

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relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic [0..*]. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem [0..*]. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship [0..*]. A list of relationships between this resource and other resources, differentiated by relationshipType.

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resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue [0..*]. A collection of resources that support this resource (bundling, link Resource Specification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.

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usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

Configuration feature.

attachment	An AttachmentRefOrValue [0..*]. Attachments relevant to this Feature, such as documentation.
featureCategory	A String [0..*]. Collection of feature categories.

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featureCharacteristic	A Characteristic [0..*]. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship [0..*]. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String [1]. The name for the feature.
note	A Note [0..*]. Notes relevant to this Feature.
policyConstraint	A PolicyRef [0..*]. Reference to manage a list of feature specification policy constraints.
relatedParty	A RelatedPartyRefOrPartyRoleRef [1..*]. Party that may be relevant to this Feature.
validFor	A CalendarPeriod [0..*]. Validity period of this Feature.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

Complements the description of an element (for instance a product) through video,

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pictures...

attachmentType	A String [1]. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, embedded inline and encoded as base64. Mutually exclusive with url.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String [1]. A technical characterization of the attachment content format using an IETF Media Type, for example application/pdf or image/png. Note: this property is named mimeType for historical reasons.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. The URL pointing to the attachment content held in an external system, such as a content management system or document repository. Mutually exclusive with content.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

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~~description~~	DEPRECATED A String. A narrative text describing the content of the attachment. DEPRECATED: will be removed in v6.
href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
~~url~~	DEPRECATED A String. Link to the attachment media/content. DEPRECATED: will be removed in v6.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String [1]. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

Another Characteristic that is related to the current Characteristic;.

id	A String [1]. Unique identifier of the characteristic.
relationshipType	A String [1]. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

A characteristic which value is an array of Boolean(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Boolean [1..*]. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

A characteristic which value is a Boolean.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Boolean [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristic sub-resource fields

A characteristic which value is an array of Float(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Float [1..*]. A characteristic which value is an array of Float(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

A characteristic which value is a Float.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Float [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	An Integer [1..*]. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

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A characteristic which value is a Integer.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	An Integer [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Number [1..*]. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

A characteristic which value is a Number.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.

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id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Number [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

A characteristic which value is an array of Object(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A valueType [1..*]. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

A characteristic which value is a Object.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A valueType_2 [1]. Value of the characteristic.

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

A characteristic which value is an array of String(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A String [1..*]. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

A characteristic which value is a String.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A String [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FeatureRelationship sub-resource fields

Configuration feature.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. This is the name of the target feature.
relationshipType	A String [1]. This is the type of the feature relationship. ENUMERATED with values: * excluded * includes * may include * requires
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

Extra information about a given entity.

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

Reference to managed Policy object.

href	A String. Hyperlink reference.
id	A String [1]. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

CalendarPeriod sub-resource fields

The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.

day	A DayOfWeekType. Possible values for the days of the week. ENUMERATED with values: * mon * tue * wed * thu * fri * sat * sun * weekdays * weekend
hourPeriod	A HourPeriod [0..*]. Start and end time range in 24-hour HH:mm format.

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status	A String [1]. Indication of the availability of the calendar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information in ISO 8601 timezone: either 'Z' for UTC or an offset like +02:00 or -05:30.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

HourPeriod sub-resource fields

Hour interval.

endHour	A String. The end of the hour period, expressed in 24-hour HH:mm format (e.g., 12:00 for noon, 23:59 for one minute before midnight).
startHour	A String. The beginning of the hour period, expressed in 24-hour HH:mm format (e.g., 09:00 for 9 AM, 17:30 for 5:30 PM).

IntentRef sub-resource fields

Intent reference, for when Intent is used by other entities.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRef sub-resource fields

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Use role to distinguish the context of the place being referenced (such as `delivery-entrance`). Use `@referredType` to distinguish the specific Place sub-type (such as a `GeographicAddress`). Use `@type` to extend (eg: as RefOrValue).

place	A PlaceRef [1]. A reference to a physical Place.
role	A String [1]. The context in which this Place is being used.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The kind of Place that is being referenced.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedResourceOrderItem sub-resource fields

RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.

itemAction	An OrderItemActionType. Action to be performed on the entity managed by the item. ENUMERATED with values: * add * modify * delete * noChange
itemId	A String. Identifier of the order item where the resource was managed.
resourceOrderHref	A String. Reference of the related entity.
resourceOrderId	A String. Unique identifier of a related entity.
role	A String. Role of the resource order item for this resource.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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ResourceRelationship sub-resource fields

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

relationshipType	A String [1]. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [relies On] if the resource needs another already owned resource to rely on (eg: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A ResourceRef [1]. Resource reference, for when Resource is used by other entities.
resourceRelationshipCharacteristic	A Characteristic [0..*]. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic... Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

Resource reference, for when Resource is used by other entities.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef sub-resource fields

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
version	A String. Resource Specification version.

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@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

CheckResourceQualificationItem sub-resource fields

A ResourceQualificationItem relates to a specific resource being checked in a qualification operation.

appointmentRequired	A Boolean. Indicates whether an appointment is required or not.
category	A ResourceCategoryRef. The (resource) category resource is used to group resource candidates in logical containers. Categories can contain other categories.
eligibilityResultReason	An EligibilityResultReason [0..**]. A list of eligibility result reasons (EligibilityResultReason [**]). Reason for eligibility result, whatever the result is (availability or unavailability).
errorMessage	An ErrorMessage [0..*]. A list of error message contains the error that caused the termination of the task.
expectedResourceAvailabilityDate	A DateTime. Date when the requester looks for resource availability.
expirationDate	A DateTime. Date when the qualification item response expires.
id	A String. Locally unique ID of the of the Resource Qualification Item.
place	A PlaceRef. A reference to a physical Place.
qualificationItemRelationship	A ResourceQualificationItemRelationship [0..*]. A list of resource qualification item relationships used to describe relationship between resourceQualification item from the same resourceQualification.

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qualificationResult	A String. Qualification result for resourceQualification item. It could be: - qualified (request resource are available), - unqualified (requested not available and not alternate available), - alternate (requested not available but proposal available).
resource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.
state	A String. State of the CheckResourceQualification: described in the state machine diagram.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

EligibilityResultReason sub-resource fields

Reason for eligibility result if the ResourceQualification result is no (meaning the Resource is not available).

code	A String. Result reason code.
label	A String. Human readable result reason label.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceQualificationItemRelationship sub-resource fields

Structure used to describe relationship between resourceQualification items from the same resourceQualification.

id	A String. Id of the resourceQualificationItem (from the same resourceQualificationItem).

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relationshipType	A String. Type of relationship (ex: reliesOn, connectedTo, etc...).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'QueryResourceQualification' resource object.


```
{
  "id": "54",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/queryResourceQualification/54",
  "description": "Query Resource Qualification for PON equipments in TeeMall",
  "effectiveQualificationDate": "2026-01-13T13:51:41.218Z",
  "estimatedResponseDate": "2026-01-13T14:00:00.218Z",
  "expectedQualificationDate": "2026-01-13T16:51:41.218Z",
  "expirationDate": "2026-01-15T17:51:41.218Z",
  "queryResourceQualificationDate": "2026-01-13T13:51:41.218Z",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v5/category/1",
      "name": "PON",
      "@type": "ResourceCategoryRef",
      "@referredType": "ResourceCategory"
    },
    "@type": "ResourceQualificationItem"
  },
  "resourceQualificationItem": [
    {
      "id": "1",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
```


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```
        "resourceSpecification": {
          "id": "78",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/78",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "9",
        "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/9",
        "name": "TeeMall 10th floor"
      },
      "@type": "ResourceQualificationItem"
    },
    {
      "id": "2",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "80",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/80",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "29",
        "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/29",
        "name": "TeeMall 30th floor"
      },
      "@type": "ResourceQualificationItem"
    }
  ],
  "state": "acknowledged",
  "@type": "QueryResourceQualification"
}
```


CheckResourceQualification resource

CheckResourceQualification is used to perform a technical eligibility on resource configuration(s). It allows to retrieve resources that are technically available in the context of the interaction (place, party, resource characteristics, ...).

Resource model

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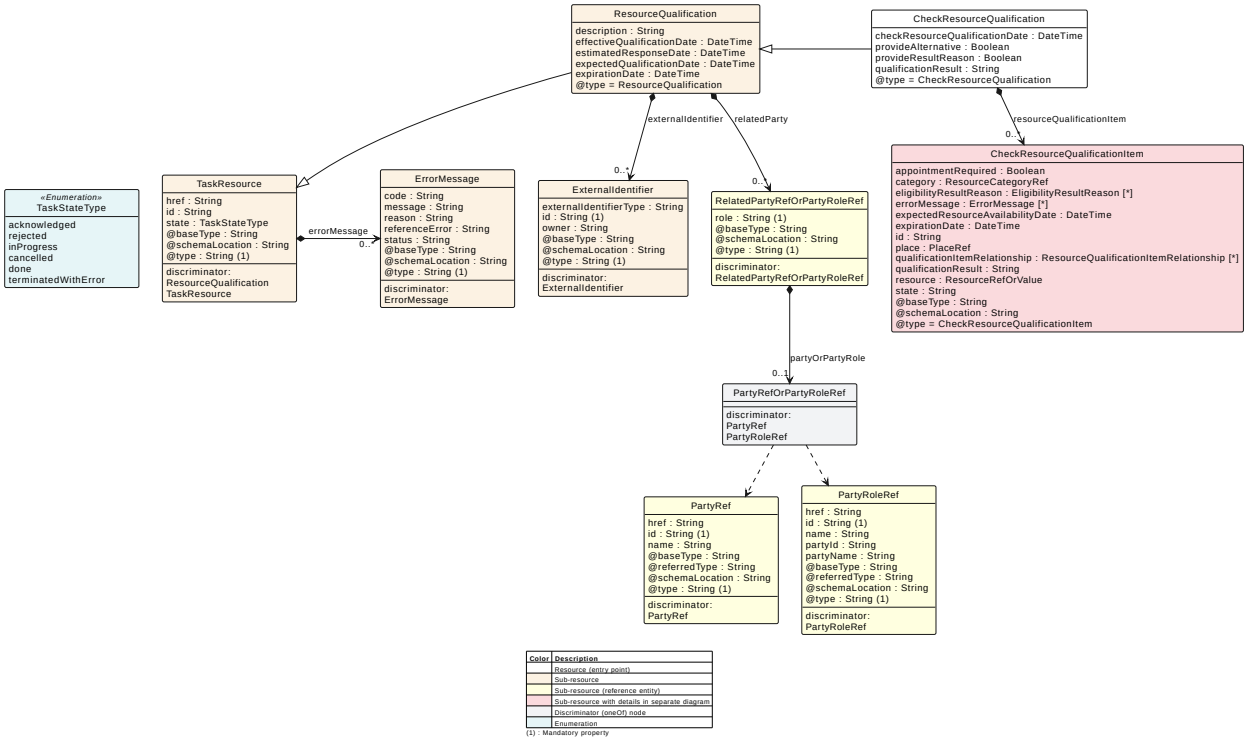


Figure 6 - CheckResourceQualification

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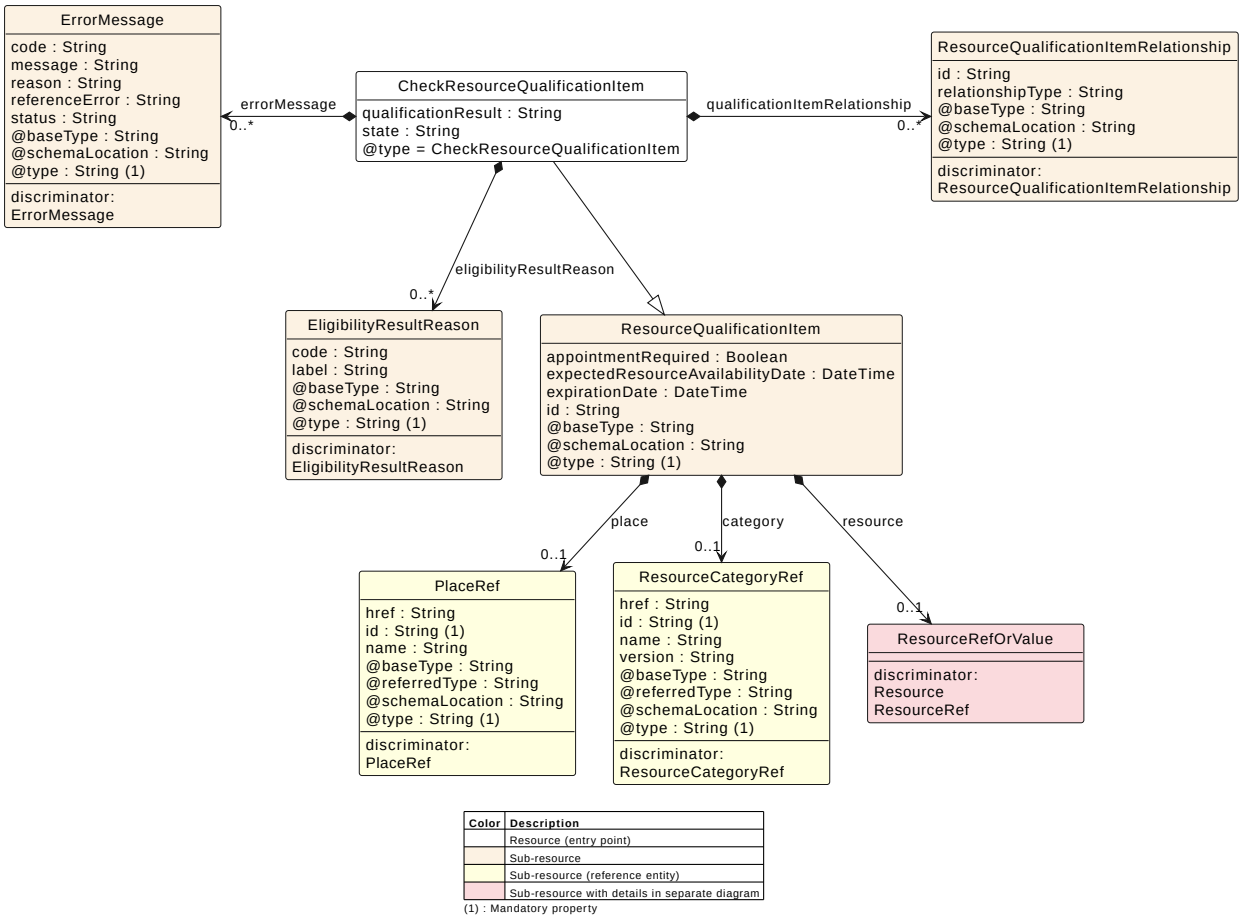


Figure 7 - CheckResourceQualificationItem

See Figure 2 for Resource diagrams.

See Figure 3 for ResourceRefOrValue diagrams.

Field descriptions

CheckResourceQualification fields

checkResourceQualificationDate	A DateTime. Date when the resourceQualification was submitted.
description	A String. Description of the resourceQualification.
effectiveQualificationDate	A DateTime. Effective date to resourceQualification completion.
errorMessage	An ErrorMessage [0..*]. A list of error message contains the error that caused the termination of the task.

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estimatedResponseDate	A DateTime. Date when the requester expect to provide an answer for the qualification request.
expectedQualificationDate	A DateTime. A date (DateTime). Deadline date when the requester expected a qualification answer.
expirationDate	A DateTime. Date when the qualification response expires.
externalIdentifier	An ExternalIdentifier [0..*]. A list of external identifiers associated with this entity.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
provideAlternative	A Boolean. When the value is TRUE means that alternative solutions should be provided.
provideResultReason	A Boolean. When the value is TRUE means that result reason are expected for available or not of resource offering.
qualificationResult	A String. Qualification result for this resourceQualification. It could be: qualified (all qualification item are qualified), alternate (At least one item alternate and no item with unqualified), unqualified (At least one item unqualified), qualifiedRestricted (all qualification item are qualified are qualified but some with a restricted set of characteristic possible values).
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of related party references, defines party or party role linked to this request.
resourceQualificationItem	A CheckResourceQualificationItem [1..*]. A list of resource qualification items.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ErrorMessage sub-resource fields

Represents an Error.

See [ErrorMessage] for details.

ExternalIdentifier sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

See [ExternalIdentifier] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

A Party reference.

See [PartyRef] for details.

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

See [PartyRoleRef] for details.

CheckResourceQualificationItem sub-resource fields

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A ResourceQualificationItem relates to a specific resource being checked in a qualification operation.

See [CheckResourceQualificationItem] for details.

ResourceCategoryRef sub-resource fields

The (resource) category resource is used to group resource candidates in logical containers. Categories can contain other categories.

See [ResourceCategoryRef] for details.

EligibilityResultReason sub-resource fields

Reason for eligibility result if the ResourceQualification result is no (meaning the Resource is not available).

See [EligibilityResultReason] for details.

PlaceRef sub-resource fields

A reference to a physical Place.

See [PlaceRef] for details.

ResourceQualificationItemRelationship sub-resource fields

Structure used to describe relationship between resourceQualification items from the same resourceQualification.

See [ResourceQualificationItemRelationship] for details.

Resource sub-resource fields

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources (e.g. TPE, EQUIPMENT) in the inventory.

See [Resource] for details.

Feature sub-resource fields

Configuration feature.

See [Feature] for details.

Attachment sub-resource fields

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Complements the description of an element (for instance a product) through video, pictures...

See [Attachment] for details.

AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

See [AttachmentRef] for details.

Characteristic sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

Another Characteristic that is related to the current Characteristic;.

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

A characteristic which value is an array of Boolean(s).

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

A characteristic which value is a Boolean.

See [BooleanCharacteristic] for details.

FloatArrayCharacteristic sub-resource fields

A characteristic which value is an array of Float(s).

See [FloatArrayCharacteristic] for details.

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FloatCharacteristic sub-resource fields

A characteristic which value is a Float.

See [FloatCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

A characteristic which value is a Integer.

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

A characteristic which value is a Number.

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

A characteristic which value is an array of Object(s).

See [ObjectArrayCharacteristic] for details.

valueType sub-resource fields

Characteristic value.

See [valueType] for details.

ObjectCharacteristic sub-resource fields

A characteristic which value is a Object.

See [ObjectCharacteristic] for details.

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valueType_2 sub-resource fields

Value of the characteristic.

See [valueType_2] for details.

StringArrayCharacteristic sub-resource fields

A characteristic which value is an array of String(s).

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

A characteristic which value is a String.

See [StringCharacteristic] for details.

FeatureRelationship sub-resource fields

Configuration feature.

See [FeatureRelationship] for details.

Note sub-resource fields

Extra information about a given entity.

See [Note] for details.

PolicyRef sub-resource fields

Reference to managed Policy object.

See [PolicyRef] for details.

CalendarPeriod sub-resource fields

The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.

See [CalendarPeriod] for details.

HourPeriod sub-resource fields

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Hour interval.

See [HourPeriod] for details.

IntentRef sub-resource fields

Intent reference, for when Intent is used by other entities.

See [IntentRef] for details.

RelatedPlaceRef sub-resource fields

Use role to distinguish the context of the place being referenced (such as `delivery-entrance`). Use `@referredType` to distinguish the specific Place sub-type (such as a `GeographicAddress`). Use `@type` to extend (eg: as RefOrValue).

See [RelatedPlaceRef] for details.

RelatedResourceOrderItem sub-resource fields

RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.

See [RelatedResourceOrderItem] for details.

ResourceRelationship sub-resource fields

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

See [ResourceRelationship] for details.

ResourceRef sub-resource fields

Resource reference, for when Resource is used by other entities.

See [ResourceRef] for details.

ResourceSpecificationRef sub-resource fields

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization

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of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

See [ResourceSpecificationRef] for details.

ResourceQualificationItem sub-resource fields

A ResourceQualificationItem relates to a specific resource being checked in a qualification operation.

See [ResourceQualificationItem] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'CheckResourceQualification' resource object.


```
{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/44",
  "checkResourceQualificationDate": "2026-01-11T19:20:39.239Z",
  "description": "check resource qualification for 1080P IPTV in the TEMALL VIP room",
  "effectiveQualificationDate": "2026-01-11T15:20:39.239Z",
  "estimatedResponseDate": "2026-01-11T15:20:39.239Z",
  "expectedQualificationDate": "2026-01-11T15:20:39.239Z",
  "expirationDate": "2026-01-24T19:20:39.239Z",
  "provideAlternative": true,
  "provideResultReason": true,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "resourceQualificationItem": [
    {
      "id": "1",
      "expectedResourceAvailabilityDate": "2026-02-11T19:20:39.239Z",
```


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```
      "expirationDate": "2026-01-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "eligibilityResultReason": [
        {
          "code": "26-001",
          "label": "not enough bandwith at this place",
          "@type": "EligibilityResultReason"
        }
      ],
      "resource": {
        "resourceCharacteristic": [
          {
            "id": "1",
            "name": "BroadcastQuality",
            "valueType": "string",
            "value": "1080P",
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic"
          }
        ],
        "resourceSpecification": {
          "id": "23",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/23",
          "name": "IPTV",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "usageState": "idle",
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "11",
        "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
        "name": "TEMALL VIP room"
      },
      "@type": "CheckResourceQualificationItem"
    }
  ],
  "state": "done",
  "@type": "CheckResourceQualification"
}
```


Notification Resource Models

7 notifications are defined for this API.

Notifications related to QueryResourceQualification:

- Create Event
- Delete Event

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- State Change Event

Notifications related to CheckResourceQualification:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

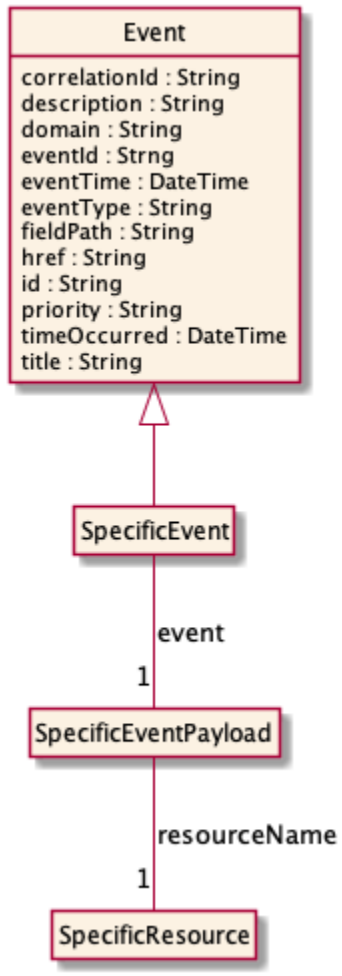


Figure 8 Notification Pattern

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QueryResourceQualification

Create Event

Message example for QueryResourceQualificationCreateEvent event

```
Content-Type: application/json
{
  "correlationId": "b122fa39-7242",
  "description": "QueryResourceQualificationCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "4d6e-98cf-a085d0121f61",
  "eventTime": "2026-01-13T18:00:00.218Z",
  "eventType": "QueryResourceQualificationCreateEvent",
  "priority": "5",
  "timeOccurred": "2026-01-13T14:00:00.218Z",
  "title": "QueryResourceQualificationCreateEvent",
  "event": {
    "queryResourceQualification": {
      "id": "54",
      "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/queryResourceQualification/54",
      "description": "Query Resource Qualification Illustration - 2 resourceSpec
available at this place for this category",
      "effectiveQualificationDate": "2026-01-13T13:51:41.218Z",
      "estimatedResponseDate": "2026-01-13T14:00:00.218Z",
      "expectedQualificationDate": "2026-01-13T16:51:41.218Z",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "externalIdentifier": [
        {
          "id": "CSQ1",
          "@type": "ExternalIdentifier"
        }
      ],
      "queryResourceQualificationDate": "2026-01-13T13:51:41.218Z",
      "relatedParty": [
        {
          "role": "Buyer contact",
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "partyOrPartyRole": {
            "id": "456",
            "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
            "name": "Tony Li",
            "@type": "PartyRef",
            "@referredType": "Individual"
          }
        }
      ],
      "searchCriteria": {
        "id": "1",
        "category": {
          "id": "1",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/category/1",
          "name": "PON",
          "@type": "ResourceCategoryRef",
          "@referredType": "ResourceCategory"
```

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```
        },
        "@type": "ResourceQualificationItem"
      },
      "resourceQualificationItem": [
        {
          "id": "1",
          "expirationDate": "2026-01-15T17:51:41.218Z",
          "resource": {
            "resourceSpecification": {
              "id": "78",
              "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/78",
              "name": "ONU",
              "@type": "ResourceSpecificationRef",
              "@referredType": "ResourceSpecification"
            },
            "@type": "Resource"
          },
          "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "9",
            "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/9",
            "name": "TeeMall 10th floor"
          },
          "@type": "ResourceQualificationItem"
        },
        {
          "id": "2",
          "expirationDate": "2026-01-15T17:51:41.218Z",
          "resource": {
            "resourceSpecification": {
              "id": "80",
              "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/80",
              "name": "ONU",
              "@type": "ResourceSpecificationRef",
              "@referredType": "ResourceSpecification"
            },
            "@type": "Resource"
          },
          "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "29",
            "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/29",
            "name": "TeeMall 30th floor"
          },
          "@type": "ResourceQualificationItem"
        }
      ],
      "state": "acknowledged",
      "@type": "QueryResourceQualification"
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
```


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```
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "QueryResourceQualificationCreateEvent"
}
```

Delete Event

Message example for QueryResourceQualificationDeleteEvent event

```
Content-Type: application/json
{
  "correlationId": "4afc7d04-5251",
  "description": "QueryResourceQualificationDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "4212-968d-a8fe5f08aa16",
  "eventTime": "2026-01-14T12:20:45.513Z",
  "eventType": "QueryResourceQualificationDeleteEvent",
  "priority": "2",
  "timeOccurred": "2026-01-14T12:20:36.791Z",
  "title": "QueryResourceQualificationDeleteEvent",
  "event": {
    "queryResourceQualification": {
      "id": "54",
      "href": "http://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/queryResourceQualification/54",
      "@type": "QueryResourceQualification"
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "QueryResourceQualificationDeleteEvent"
}
```

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State Change Event

Message example for QueryResourceQualificationStateChangeEvent event

```
Content-Type: application/json
{
  "correlationId": "95e2f5d5-e67f",
  "description": "QueryResourceQualificationStateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "4011-abc1-8a9dfc8ff694",
  "eventTime": "2026-01-13T14:30:00.218Z",
  "eventType": "QueryResourceQualificationStateChangeEvent",
  "priority": "5",
  "timeOccurred": "2026-01-13T14:30:00.218Z",
  "title": "QueryResourceQualificationStateChangeEvent",
  "event": {
    "queryResourceQualification": {
      "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/queryResourceQualification/54",
      "id": "54",
      "@type": "QueryResourceQualification",
      "state": "done"
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "QueryResourceQualificationStateChangeEvent"
}
```

CheckResourceQualification

Create Event

Message example for CheckResourceQualificationCreateEvent event

```
Content-Type: application/json
{
  "correlationId": "901e0ff8-b717",
  "description": "CheckResourceQualificationCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "439e-8ffd-25b84f13b3df",
  "eventTime": "2026-01-11T15:20:39.239Z",
  "eventType": "CheckResourceQualificationCreateEvent",
```

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```
  "priority": "2",
  "timeOccurred": "2026-01-11T15:20:39.239Z",
  "title": "CheckResourceQualificationCreateEvent",
  "event": {
    "checkResourceQualification": {
      "id": "44",
      "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/44",
      "checkResourceQualificationDate": "2026-01-11T19:20:39.239Z",
      "description": "check resource qualification illustration",
      "effectiveQualificationDate": "2026-01-11T15:20:39.239Z",
      "estimatedResponseDate": "2026-01-11T15:20:39.239Z",
      "expectedQualificationDate": "2026-01-11T15:20:39.239Z",
      "expirationDate": "2026-01-24T19:20:39.239Z",
      "externalIdentifier": [
        {
          "id": "CSQ1",
          "@type": "ExternalIdentifier"
        }
      ],
      "provideAlternative": true,
      "provideResultReason": true,
      "qualificationResult": "alternate",
      "relatedParty": [
        {
          "role": "Buyer contact",
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "partyOrPartyRole": {
            "id": "456",
            "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
            "name": "Tony Li",
            "@type": "PartyRef",
            "@referredType": "Individual"
          }
        }
      ],
      "resourceQualificationItem": [
        {
          "id": "1",
          "expectedResourceAvailabilityDate": "2026-02-11T19:20:39.239Z",
          "expirationDate": "2026-01-24T19:20:39.239Z",
          "qualificationResult": "alternate",
          "state": "done",
          "eligibilityResultReason": [
            {
              "code": "26-001",
              "label": "not enough bandwith at this place",
              "@type": "EligibilityResultReason"
            }
          ],
          "resource": {
            "resourceCharacteristic": [
              {
                "id": "1",
                "name": "BroadcastQuality",
                "valueType": "string",
                "value": "1080P",
                "@type": "StringCharacteristic"
              }
            ],
```


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```
            "resourceSpecification": {
              "id": "23",
              "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/23",
              "name": "IPTV",
              "@type": "ResourceSpecificationRef",
              "@referredType": "ResourceSpecification"
            },
            "usageState": "busy",
            "@type": "Resource"
          },
          "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "11",
            "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
            "name": "TEMALL VIP room"
          },
          "@type": "CheckResourceQualificationItem"
        }
      ],
      "state": "done",
      "@type": "CheckResourceQualification"
    }
  },
  "reportingSystem": {
    "id": "954",
    "name": "APP-190",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "441",
    "name": "APP-643",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "CheckResourceQualificationCreateEvent"
}
```


Delete Event

Message example for CheckResourceQualificationDeleteEvent event


```
Content-Type: application/json
{
  "correlationId": "b3727ad0-1140",
  "description": "CheckResourceQualificationDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "4c58-840e-790e0657230f",
  "eventTime": "2026-01-10T12:41:51.932Z",
  "eventType": "CheckResourceQualificationDeleteEvent",
  "priority": "4",
  "timeOccurred": "2023-03-05T12:41:46.084Z",
```


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```
  "title": "CheckResourceQualificationDeleteEvent",
  "event": {
    "checkResourceQualification": {
      "id": "456",
      "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/456",
      "@type": "CheckResourceQualification"
    }
  },
  "reportingSystem": {
    "id": "954",
    "name": "APP-190",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "441",
    "name": "APP-643",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "CheckResourceQualificationDeleteEvent"
}
```


Attribute Value Change Event

Message example for CheckResourceQualificationAttributeValueChangeEvent event


```
Content-Type: application/json
{
  "correlationId": "e7e44c9a-64f5",
  "description": "CheckResourceQualificationAttributeValueChangeEvent illustration",
  "domain": "Resource",
  "eventId": "483f-800d-f34613c8d3e2",
  "eventTime": "2026-01-15T12:20:45.443Z",
  "eventType": "CheckResourceQualificationAttributeValueChangeEvent",
  "priority": "1",
  "timeOccurred": "2026-01-16T12:20:42.512Z",
  "title": "CheckResourceQualificationAttributeValueChangeEvent",
  "event": {
    "checkResourceQualification": {
      "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/44",
      "id": "44",
      "@type": "CheckResourceQualification",
      "relatedParty": [
        {
          "role": "Buyer contact",
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "partyOrPartyRole": {
            "id": "456",
            "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
            "name": "Tony Li",
            "@type": "PartyRef",
            "@referredType": "Individual"
```


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```
          }
        }
      ]
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "CheckResourceQualificationAttributeValueChangeEvent"
}
```


State Change Event

Message example for CheckResourceQualificationStateChangeEvent event with ?fields=state property only


```
Content-Type: application/json
{
  "correlationId": "a19c7353-eae1",
  "description": "CheckResourceQualificationStateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "45a4-81bb-85fa6cc903c8",
  "eventTime": "2026-01-11T15:30:39.239Z",
  "eventType": "CheckResourceQualificationStateChangeEvent",
  "priority": "5",
  "timeOccurred": "2026-01-11T15:30:39.239Z",
  "title": "CheckResourceQualificationStateChangeEvent",
  "event": {
    "checkResourceQualification": {
      "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/44",
      "id": "44",
      "@type": "CheckResourceQualification",
      "state": "done"
    }
  },
  "reportingSystem": {
    "id": "954",
    "name": "APP-190",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "441",
    "name": "APP-643",
    "@type": "ReportingResource",
```


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```
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "CheckResourceQualificationStateChangeEvent"
  }
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

> - Filtering and attribute selection rules are described in the TMF REST Design Guidelines.
> - Since API requests are asynchronous and follow the TMF task-driven design pattern, there is a task execution period from when the API is actually triggered until it returns data. During this execution time, users can modify or delete the original request.

Notifications are also described in a subsequent section.

Operations on QueryResourceQualification

Retrieves a queryResourceQualification by ID

`GET /queryResourceQualification/{id}?fields=…`

Description

This operation retrieves a QueryResourceQualification entity. Attribute selection is enabled

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for all first level attributes.

List or find queryResourceQualification objects

`GET /queryResourceQualification?fields=…&{filtering}`

Description

This operation list QueryResourceQualification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of query resource qualification request(s). The given criteria is the query resource qualification state (done).

Request

```
GET /queryResourceQualification?filter=state=done
Content-Type: application/json
```

Response

```
200

[
  {
    "id": "54",
    "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/queryResourceQualification/54",
    "description": "Query Resource Qualification Illustration - 2 resourceSpec
available at this place for this category",
    "effectiveQualificationDate": "2026-01-13T13:51:41.218Z",
    "estimatedResponseDate": "2026-01-13T14:00:00.218Z",
    "expectedQualificationDate": "2026-01-13T16:51:41.218Z",
    "expirationDate": "2026-01-15T17:51:41.218Z",
    "externalIdentifier": [
      {
        "id": "CSQ1",
        "@type": "ExternalIdentifier"
      }
    ],
    "queryResourceQualificationDate": "2026-01-13T13:51:41.218Z",
    "relatedParty": [
      {
        "role": "Buyer contact",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "id": "456",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
```

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```
          "name": "Tony Li",
          "@type": "PartyRef",
          "@referredType": "Individual"
        }
      }
    ],
    "searchCriteria": {
      "id": "1",
      "category": {
        "id": "1",
        "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/category/1",
        "name": "PON",
        "@type": "ResourceCategoryRef",
        "@referredType": "ResourceCategory"
      },
      "@type": "ResourceQualificationItem"
    },
    "resourceQualificationItem": [
      {
        "id": "1",
        "expirationDate": "2026-01-15T17:51:41.218Z",
        "resource": {
          "resourceSpecification": {
            "id": "78",
            "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/78",
            "name": "ONU",
            "@type": "ResourceSpecificationRef",
            "@referredType": "ResourceSpecification"
          },
          "@type": "Resource"
        },
        "place": {
          "@type": "PlaceRef",
          "@referredType": "GeographicSite",
          "id": "9",
          "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/9",
          "name": "TeeMall 10th floor"
        },
        "@type": "ResourceQualificationItem"
      },
      {
        "id": "2",
        "expirationDate": "2026-01-15T17:51:41.218Z",
        "resource": {
          "resourceSpecification": {
            "id": "80",
            "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/80",
            "name": "ONU",
            "@type": "ResourceSpecificationRef",
            "@referredType": "ResourceSpecification"
          },
          "@type": "Resource"
        },
        "place": {
          "@type": "PlaceRef",
          "@referredType": "GeographicSite",
```


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```
          "id": "29",
          "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/29",
          "name": "TeeMall 30th floor"
        },
        "@type": "ResourceQualificationItem"
      }
    ],
    "state": "acknowledged",
    "@type": "QueryResourceQualification"
  }
]
```

Creates a queryResourceQualification

`POST /queryResourceQualification?fields=…&{filtering}`

Description

This operation creates a QueryResourceQualification entity.

Mandatory Attributes

Mandatory Attributes	Rule
searchCriteria	
@type	

Usage samples

Here's an example of a request for creating QueryResourceQualification.

Request

```
POST /queryResourceQualification
Content-Type: application/json

{
  "description": "Query Resource Qualification POST Illustration ",
  "expectedQualificationDate": "2026-01-08T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
```

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```
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v5/category/1",
      "name": "PON",
      "@type": "ResourceCategoryRef",
      "@referredType": "ResourceCategory"
    },
    "resource": {
      "lifecycleState": "planned",
      "resourceSpecification": {
        "id": "78",
        "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/78",
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourceSpecification"
      },
      "@type": "Resource"
    },
    "place": {
      "@type": "PlaceRef",
      "@referredType": "GeographicSite",
      "id": "11",
      "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/11",
      "name": "TeeMall, Tianhe District, Guangzhou"
    },
    "@type": "ResourceQualificationItem"
  },
  "@type": "QueryResourceQualification"
}
```


Response


```
201

{
  "id": "54",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/queryResourceQualification/54",
  "description": "Query Resource Qualification Illustration - 2 resourceSpec available
at this place for this category",
  "effectiveQualificationDate": "2026-01-13T13:51:41.218Z",
  "estimatedResponseDate": "2026-01-13T14:00:00.218Z",
  "expectedQualificationDate": "2026-01-13T16:51:41.218Z",
  "expirationDate": "2026-01-15T17:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
```


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```
      "@type": "ExternalIdentifier"
    }
  ],
  "queryResourceQualificationDate": "2026-01-13T13:51:41.218Z",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v5/category/1",
      "name": "PON",
      "@type": "ResourceCategoryRef",
      "@referredType": "ResourceCategory"
    },
    "@type": "ResourceQualificationItem"
  },
  "resourceQualificationItem": [
    {
      "id": "1",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "78",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/78",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "9",
        "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/9",
        "name": "TeeMall 10th floor"
      },
      "@type": "ResourceQualificationItem"
    },
    {
      "id": "2",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "80",
          "href": "https://mycsp.com:8080/tmf-
```


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```
 api/resourceCatalogManagement/v5/resourceSpecification/80",
           "name": "ONU",
           "@type": "ResourceSpecificationRef",
           "@referredType": "ResourceSpecification"
         },
         "@type": "Resource"
       },
       "place": {
         "@type": "PlaceRef",
         "@referredType": "GeographicSite",
         "id": "29",
         "href":
 "http://serverlocation:port/geographicSiteManagement/v5/geographicSite/29",
         "name": "TeeMall 30th floor"
       },
       "@type": "ResourceQualificationItem"
     }
   ],
   "state": "acknowledged",
   "@type": "QueryResourceQualification"
 }
```


Updates partially a queryResourceQualification

`PATCH /queryResourceQualification/{id}?fields=…`

Description

This operation allows partial updates of a QueryResourceQualification entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
description	

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Patchable Attributes	Rule
effectiveQualificationDate	
errorMessage	
estimatedResponseDate	
expectedQualificationDate	
expirationDate	
externalIdentifier	
queryResourceQualificationDate	
relatedParty	
resourceQualificationItem	
searchCriteria	
state	

Usage samples

Here's an example of a request for partial update of QueryResourceQualification using merge-patch+json with change on following attributes - externalIdentifier and expectedQualificationDate value.

Request

```
PATCH /queryResourceQualification/54
Content-Type: application/merge-patch+json

{
  "expectedQualificationDate": "2026-01-17T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "@type": "QueryResourceQualification"
}
```

Response

```
200

{
```

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```
  "id": "54",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/queryResourceQualification/54",
  "description": "Query Resource Qualification Illustration - 2 resourceSpec available
at this place for this category",
  "effectiveQualificationDate": "2026-01-13T13:51:41.218Z",
  "estimatedResponseDate": "2026-01-13T14:00:00.218Z",
  "expectedQualificationDate": "2026-01-17T13:51:41.218Z",
  "expirationDate": "2026-01-15T17:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "queryResourceQualificationDate": "2026-01-13T13:51:41.218Z",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v5/category/1",
      "name": "PON",
      "@type": "ResourceCategoryRef",
      "@referredType": "ResourceCategory"
    },
    "@type": "ResourceQualificationItem"
  },
  "resourceQualificationItem": [
    {
      "id": "1",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "78",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/78",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "9",
        "href":
```


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```
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/9",
        "name": "TeeMall 10th floor"
      },
      "@type": "ResourceQualificationItem"
    },
    {
      "id": "2",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "80",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/80",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "29",
        "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/29",
        "name": "TeeMall 30th floor"
      },
      "@type": "ResourceQualificationItem"
    }
  ],
  "state": "acknowledged",
  "@type": "QueryResourceQualification"
}
```


Here's an example of a request for partial update of QueryResourceQualification using json-patch+json with change on expectedQualificationDate value .

Request

```
PATCH /queryResourceQualification/54
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "expectedQualificationDate",
    "value": "2026-01-17T13:51:41.218Z"
  }
]
```

Response

```
200
```

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```
{
  "id": "54",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/queryResourceQualification/54",
  "description": "Query Resource Qualification Illustration - 2 resourceSpec available
at this place for this category",
  "effectiveQualificationDate": "2026-01-13T13:51:41.218Z",
  "estimatedResponseDate": "2026-01-13T14:00:00.218Z",
  "expectedQualificationDate": "2026-01-17T13:51:41.218Z",
  "expirationDate": "2026-01-15T17:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "queryResourceQualificationDate": "2026-01-13T13:51:41.218Z",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v5/category/1",
      "name": "PON",
      "@type": "ResourceCategoryRef",
      "@referredType": "ResourceCategory"
    },
    "@type": "ResourceQualificationItem"
  },
  "resourceQualificationItem": [
    {
      "id": "1",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "78",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/78",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "9",
```


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```
        "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/9",
        "name": "TeeMall 10th floor"
      },
      "@type": "ResourceQualificationItem"
    },
    {
      "id": "2",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "80",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/80",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "29",
        "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/29",
        "name": "TeeMall 30th floor"
      },
      "@type": "ResourceQualificationItem"
    }
  ],
  "state": "acknowledged",
  "@type": "QueryResourceQualification"
}
```


Here's an example of a request for partial update of QueryResourceQualification using json-patch-query with change on relatedParty value.

Request

```
PATCH /queryResourceQualification/54
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path": "/relatedParty/partyOrPartyRole/name?relatedParty.partyOrPartyRole.id=678",
    "value": "Ross"
  }
]
```

Response

```
200
```

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```
{
  "id": "54",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/queryResourceQualification/54",
  "description": "Query Resource Qualification Illustration - 2 resourceSpec available
at this place for this category",
  "effectiveQualificationDate": "2026-01-13T13:51:41.218Z",
  "estimatedResponseDate": "2026-01-13T14:00:00.218Z",
  "expectedQualificationDate": "2026-01-17T13:51:41.218Z",
  "expirationDate": "2026-01-15T17:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "queryResourceQualificationDate": "2026-01-13T13:51:41.218Z",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "678",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Ross",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v5/category/1",
      "name": "PON",
      "@type": "ResourceCategoryRef",
      "@referredType": "ResourceCategory"
    },
    "@type": "ResourceQualificationItem"
  },
  "resourceQualificationItem": [
    {
      "id": "1",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "78",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/78",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
```


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```
        "id": "9",
        "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/9",
        "name": "TeeMall 10th floor"
      },
      "@type": "ResourceQualificationItem"
    },
    {
      "id": "2",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "80",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/80",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "29",
        "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/29",
        "name": "TeeMall 30th floor"
      },
      "@type": "ResourceQualificationItem"
    }
  ],
  "state": "acknowledged",
  "@type": "QueryResourceQualification"
}
```


Here's an example of a request for partial update of QueryResourceQualification with change on expectedQualificationDate, externalIdentifier, relatedParty value.

Request

```
PATCH /queryResourceQualification/54
Content-Type: application/json

{
  "expectedQualificationDate": "2026-01-17T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "@type": "QueryResourceQualification"
}
```

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Response


```
200

{
  "id": "54",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/queryResourceQualification/54",
  "description": "Query Resource Qualification Illustration - 2 resourceSpec available
at this place for this category",
  "effectiveQualificationDate": "2026-01-13T13:51:41.218Z",
  "estimatedResponseDate": "2026-01-13T14:00:00.218Z",
  "expectedQualificationDate": "2026-01-17T13:51:41.218Z",
  "expirationDate": "2026-01-15T17:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "queryResourceQualificationDate": "2026-01-13T13:51:41.218Z",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "678",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Ross",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v5/category/1",
      "name": "PON",
      "@type": "ResourceCategoryRef",
      "@referredType": "ResourceCategory"
    },
    "@type": "ResourceQualificationItem"
  },
  "resourceQualificationItem": [
    {
      "id": "1",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "78",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/78",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
```


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```
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "9",
        "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/9",
        "name": "TeeMall 10th floor"
      },
      "@type": "ResourceQualificationItem"
    },
    {
      "id": "2",
      "expirationDate": "2026-01-15T17:51:41.218Z",
      "resource": {
        "resourceSpecification": {
          "id": "80",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/80",
          "name": "ONU",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "29",
        "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/29",
        "name": "TeeMall 30th floor"
      },
      "@type": "ResourceQualificationItem"
    }
  ],
  "state": "acknowledged",
  "@type": "QueryResourceQualification"
}
```


Deletes a queryResourceQualification

`DELETE /queryResourceQualification/{id}?{filtering}`

Description

This operation deletes a QueryResourceQualification entity.

Usage samples

Here's an example of a request for deleting a specific QueryResourceQualification instance.

Request

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```
DELETE /queryResourceQualification/54
Content-Type: application/json
```

Response

```
204
```

Operations on CheckResourceQualification

Retrieves a checkResourceQualification by ID

`GET /checkResourceQualification/{id}?fields=…`

Description

This operation retrieves a CheckResourceQualification entity. Attribute selection is enabled for all first level attributes.

List or find checkResourceQualification objects

`GET /checkResourceQualification?fields=…&{filtering}`

Description

This operation list CheckResourceQualification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving all the CheckResourceQualification that were completed.

Request

```
GET /checkResourceQualification?filter=state=done&fields=id,name,state
Content-Type: application/json
```

Response

```
200

[
```

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```
  {
    "id": "44",
    "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/44",
    "checkResourceQualificationDate": "2026-01-11T19:20:39.239Z",
    "description": "check resource qualification illustration",
    "effectiveQualificationDate": "2026-01-11T15:20:39.239Z",
    "estimatedResponseDate": "2026-01-11T15:20:39.239Z",
    "expectedQualificationDate": "2026-01-11T15:20:39.239Z",
    "expirationDate": "2026-01-24T19:20:39.239Z",
    "externalIdentifier": [
      {
        "id": "CSQ1",
        "@type": "ExternalIdentifier"
      }
    ],
    "provideAlternative": true,
    "provideResultReason": true,
    "qualificationResult": "alternate",
    "relatedParty": [
      {
        "role": "Buyer contact",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "id": "456",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
          "name": "Tony Li",
          "@type": "PartyRef",
          "@referredType": "Individual"
        }
      }
    ],
    "resourceQualificationItem": [
      {
        "id": "1",
        "expectedResourceAvailabilityDate": "2026-02-11T19:20:39.239Z",
        "expirationDate": "2026-01-24T19:20:39.239Z",
        "qualificationResult": "alternate",
        "state": "done",
        "eligibilityResultReason": [
          {
            "code": "26-001",
            "label": "not enough bandwith at this place",
            "@type": "EligibilityResultReason"
          }
        ],
        "resource": {
          "resourceCharacteristic": [
            {
              "id": "1",
              "name": "BroadcastQuality",
              "valueType": "string",
              "value": "1080P",
              "@type": "StringCharacteristic"
            }
          ],
          "resourceSpecification": {
            "id": "23",
            "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/23",
```


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```
            "name": "IPTV",
            "@type": "ResourceSpecificationRef",
            "@referredType": "ResourceSpecification"
          },
          "usageState": "busy",
          "@type": "Resource"
        },
        "place": {
          "@type": "PlaceRef",
          "@referredType": "GeographicSite",
          "id": "11",
          "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
          "name": "TEMALL VIP room"
        },
        "@type": "CheckResourceQualificationItem"
      }
    ],
    "state": "done",
    "@type": "CheckResourceQualification"
  }
]
```

Creates a checkResourceQualification

`POST /checkResourceQualification?fields=…&{filtering}`

Description

This operation creates a CheckResourceQualification entity.

Mandatory Attributes

Mandatory Attributes	Rule
resourceQualificationItem	
@type	

Usage samples

Here's an example of a request for creating CheckResourceQualification.

Request

```
POST /checkResourceQualification
Content-Type: application/json

{
  "description": "check resource qualification illustration",
  "expectedQualificationDate": "2026-01-11T15:20:39.239Z",
  "externalIdentifier": [
    {
```

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```
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "resourceQualificationItem": [
    {
      "id": "1",
      "expectedResourceAvailabilityDate": "2026-01-11T19:20:39.239Z",
      "resource": {
        "resourceCharacteristic": [
          {
            "id": "1",
            "name": "BroadcastQuality",
            "valueType": "string",
            "value": "1080P",
            "@type": "StringCharacteristic"
          }
        ],
        "resourceSpecification": {
          "id": "23",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/23",
          "name": "IPTV",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "usageState": "busy",
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "11",
        "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
        "name": "TEMALL VIP room"
      },
      "@type": "CheckResourceQualificationItem"
    }
  ],
  "@type": "CheckResouceQualification"
}
```


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Response

```
201

{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/44",
  "checkResourceQualificationDate": "2026-01-11T19:20:39.239Z",
  "description": "check resource qualification illustration",
  "effectiveQualificationDate": "2026-01-11T15:20:39.239Z",
  "estimatedResponseDate": "2026-01-11T15:20:39.239Z",
  "expectedQualificationDate": "2026-01-11T15:20:39.239Z",
  "expirationDate": "2026-01-24T19:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "resourceQualificationItem": [
    {
      "id": "1",
      "expectedResourceAvailabilityDate": "2026-02-11T19:20:39.239Z",
      "expirationDate": "2026-01-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "eligibilityResultReason": [
        {
          "code": "26-001",
          "label": "not enough bandwith at this place",
          "@type": "EligibilityResultReason"
        }
      ],
      "resource": {
        "resourceCharacteristic": [
          {
            "id": "1",
            "name": "BroadcastQuality",
            "valueType": "string",
            "value": "1080P",
            "@type": "StringCharacteristic"
```

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```
          }
        ],
        "resourceSpecification": {
          "id": "23",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/23",
          "name": "IPTV",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "usageState": "busy",
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "11",
        "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
        "name": "TEMALL VIP room"
      },
      "@type": "CheckResourceQualificationItem"
    }
  ],
  "state": "done",
  "@type": "CheckResourceQualification"
}
```


Updates partially a checkResourceQualification

`PATCH /checkResourceQualification/{id}?fields=…`

Description

This operation allows partial updates of a CheckResourceQualification entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable

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Non Patchable Attributes	Rule
@type	@type is immutable

Patchable Attributes	Rule
checkResourceQualificationDate	
description	
effectiveQualificationDate	
errorMessage	
estimatedResponseDate	
expectedQualificationDate	
expirationDate	
externalIdentifier	
provideAlternative	
provideResultReason	
qualificationResult	
relatedParty	
resourceQualificationItem	
state	

Usage samples

Here's an example of a request for patching a CheckResourceQualification resource (created via previous POST) with changes on attributes- externalIdentifier and expectedQualificationDate using merge-patch+json.

Request

```
PATCH /checkResourceQualification/44
Content-Type: application/merge-patch+json

{
  "expectedQualificationDate": "2026-01-14T15:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
```

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```
  "@type": "CheckResourceQualification"
}
```

Response

```
200

{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/44",
  "checkResourceQualificationDate": "2026-01-11T19:20:39.239Z",
  "description": "check resource qualification illustration",
  "effectiveQualificationDate": "2026-01-11T15:20:39.239Z",
  "estimatedResponseDate": "2026-01-11T15:20:39.239Z",
  "expectedQualificationDate": "2026-01-14T15:20:39.239Z",
  "expirationDate": "2026-01-24T19:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "resourceQualificationItem": [
    {
      "id": "1",
      "expectedResourceAvailabilityDate": "2026-02-11T19:20:39.239Z",
      "expirationDate": "2026-01-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "eligibilityResultReason": [
        {
          "code": "26-001",
          "label": "not enough bandwith at this place",
          "@type": "EligibilityResultReason"
        }
      ],
      "resource": {
        "resourceCharacteristic": [
          {
            "id": "1",
```

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```
            "name": "BroadcastQuality-4K",
            "valueType": "string",
            "value": "1080P",
            "@type": "StringCharacteristic"
          }
        ],
        "resourceSpecification": {
          "id": "23",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/23",
          "name": "IPTV",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "usageState": "busy",
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "11",
        "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
        "name": "TEMALL VIP room"
      },
      "@type": "CheckResourceQualificationItem"
    }
  ],
  "state": "done",
  "@type": "CheckResourceQualification"
}
```


Here's an example of a request for patching a CheckResourceQualification resource (created via previous POST) with updated expectedQualificationDate using json-patch+json.

Request

```
PATCH /checkResourceQualification/44
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "expectedQualificationDate",
    "value": "2026-01-16T15:20:39.239Z"
  }
]
```

Response

```
200
```

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```
{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/44",
  "checkResourceQualificationDate": "2026-01-11T19:20:39.239Z",
  "description": "check resource qualification illustration",
  "effectiveQualificationDate": "2026-01-11T15:20:39.239Z",
  "estimatedResponseDate": "2026-01-11T15:20:39.239Z",
  "expectedQualificationDate": "2026-01-16T15:20:39.239Z",
  "expirationDate": "2026-01-24T19:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "resourceQualificationItem": [
    {
      "id": "1",
      "expectedResourceAvailabilityDate": "2026-02-11T19:20:39.239Z",
      "expirationDate": "2026-01-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "eligibilityResultReason": [
        {
          "code": "26-001",
          "label": "not enough bandwith at this place",
          "@type": "EligibilityResultReason"
        }
      ],
      "resource": {
        "resourceCharacteristic": [
          {
            "id": "1",
            "name": "BroadcastQuality",
            "valueType": "string",
            "value": "1080P",
            "@type": "StringCharacteristic"
          }
        ],
        "resourceSpecification": {
          "id": "23",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/23",
```


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```
          "name": "IPTV",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "usageState": "busy",
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "11",
        "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
        "name": "TEMALL VIP room"
      },
      "@type": "CheckResourceQualificationItem"
    }
  ],
  "state": "done",
  "@type": "CheckResourceQualification"
}
```

Here's an example of a request for patching a CheckResourceQualification resource (created via previous POST) with change on expectedActivationDate using json-patch-query.

Request

```
PATCH /checkResourceQualification/44
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path":
"/resourceQualificationItem/expectedActivationDate?resourceQualificationItem.id=1",
    "value": "2026-01-01T19:20:39.239Z"
  }
]
```

Response

```
200

{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/44",
  "checkResourceQualificationDate": "2026-01-11T19:20:39.239Z",
  "description": "check resource qualification illustration",
  "effectiveQualificationDate": "2026-01-11T15:20:39.239Z",
  "estimatedResponseDate": "2026-01-11T15:20:39.239Z",
  "expectedQualificationDate": "2026-01-16T15:20:39.239Z",
```

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```
  "expirationDate": "2026-01-24T19:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "resourceQualificationItem": [
    {
      "id": "1",
      "expectedResourceAvailabilityDate": "2026-02-11T19:20:39.239Z",
      "expirationDate": "2026-01-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "eligibilityResultReason": [
        {
          "code": "26-001",
          "label": "not enough bandwith at this place",
          "@type": "EligibilityResultReason"
        }
      ],
      "resource": {
        "resourceCharacteristic": [
          {
            "id": "1",
            "name": "BroadcastQuality",
            "valueType": "string",
            "value": "1080P",
            "@type": "StringCharacteristic"
          }
        ],
        "resourceSpecification": {
          "id": "23",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/23",
          "name": "IPTV",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "usageState": "busy",
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
```


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```
        "@referredType": "GeographicSite",
        "id": "11",
        "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
        "name": "TEMALL VIP room"
      },
      "@type": "CheckResourceQualificationItem"
    }
  ],
  "state": "done",
  "@type": "CheckResourceQualification"
}
```


Here's an example of a request for patching a CheckResourceQualification resource (created via previous POST) with change on attributes- externalIdentifier and expectedQualificationDate value

Request

```
PATCH /checkResourceQualification/44
Content-Type: application/json

{
  "expectedQualificationDate": "2026-01-17T15:23:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-002",
      "@type": "ExternalIdentifier"
    }
  ],
  "@type": "CheckResourceQualification"
}
```

Response


```
200

{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-
api/resourceQualificationManagement/v5/checkResourceQualification/44",
  "checkResourceQualificationDate": "2026-01-11T19:20:39.239Z",
  "description": "check resource qualification illustration",
  "effectiveQualificationDate": "2026-01-11T15:20:39.239Z",
  "estimatedResponseDate": "2026-01-11T15:20:39.239Z",
  "expectedQualificationDate": "2026-01-17T15:23:39.239Z",
  "expirationDate": "2026-01-24T19:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-002",
      "@type": "ExternalIdentifier"
    }
  ],
```


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```
  "provideAlternative": true,
  "provideResultReason": true,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Tony Li",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "resourceQualificationItem": [
    {
      "id": "1",
      "expectedResourceAvailabilityDate": "2026-02-11T19:20:39.239Z",
      "expirationDate": "2026-01-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "eligibilityResultReason": [
        {
          "code": "26-001",
          "label": "not enough bandwith at this place",
          "@type": "EligibilityResultReason"
        }
      ],
      "resource": {
        "resourceCharacteristic": [
          {
            "id": "1",
            "name": "BroadcastQuality",
            "valueType": "string",
            "value": "1080P",
            "@type": "StringCharacteristic"
          }
        ],
        "resourceSpecification": {
          "id": "23",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/23",
          "name": "IPTV",
          "@type": "ResourceSpecificationRef",
          "@referredType": "ResourceSpecification"
        },
        "usageState": "busy",
        "@type": "Resource"
      },
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "11",
        "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
        "name": "TEMALL VIP room"
      },
      "@type": "CheckResourceQualificationItem"
```


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```
    }
  ],
  "state": "done",
  "@type": "CheckResourceQualification"
}
```

Deletes a checkResourceQualification

`DELETE /checkResourceQualification/{id}?{filtering}`

Description

This operation deletes a CheckResourceQualification entity.

Usage samples

Here's an example of a request for deleting a specific CheckResourceQualification instance.

Request

```
DELETE /checkResourceQualification/456
Content-Type: application/json
```

Response

```
204

{
    "id": "456",
    "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/checkServiceQualification/456",
    "@type": "CheckResourceQualification"
}
```

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42
```

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```
{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

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Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Version 5.0.0	07/01/2026	Tony Lee, China Telecom, lisy35@chinatelecom.cn	(Initial version)

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